

Ramanuj Saket

AI/ML/Gen-AI Engineer

📞 +91-8349819726 ✉️ rmnjsaket4664@gmail.com 🔗 LinkedIn Profile 🐙 Github Profile
📍 Mandi, Himachal Pradesh, India

Professional Synopsis

AI/ML Engineer with 4+ years of experience, including 3+ years focused on Generative AI, Agentic AI, and machine learning model development. Currently working at IIT Mandi iHub & HCI Foundation, building and deploying AI systems for healthcare, travel, and industrial clients. Pursuing an Executive M.Tech in Artificial Intelligence at IIT Jodhpur alongside full-time work. Experienced in building AI chatbots, computer vision systems, and automated agent workflows, and in deploying models on AWS cloud infrastructure. Comfortable working directly with clients and cross-functional teams, and known for clear technical documentation and mentoring junior team members.

Core Technical Skills

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Neural Networks, Feature Engineering (imputation, outlier handling, binning, normalization), Regularization (L1/L2), TensorFlow, PyTorch, Scikit-learn, XGBoost
Generative AI & Agentic AI: LLMs, Retrieval-Augmented Generation (RAG), LangChain, Semantic Kernel, AutoGen, CrewAI, Prompt Engineering, LLM Fine-Tuning (LoRA, QLoRA), N8N, Microsoft Agent Framework
Cloud & MLOps: AWS SageMaker, AWS Bedrock, EC2, Lightsail, Lambda, DynamoDB, S3, Rekognition, Textract, Cloudflare AI, RunPod, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD
Computer Vision & NLP: OpenCV, YOLOv8, DeepFace, MediaPipe, MTCNN, spaCy
Programming & Databases: Python, MySQL, MongoDB, PostgreSQL, FAISS Vector DB
Data Analysis & Visualization: Power BI, NumPy, Pandas, Matplotlib

Professional Experience

IIT Mandi iHub & HCI Foundation

Aug 2023 – Present

Software Developer

Mandi, H.P.

- Designed, trained, and deployed production-ready Machine Learning and Generative AI models, covering data preprocessing, model fine-tuning, and performance optimization.
- Built and integrated intelligent, agent-based AI systems using frameworks such as Semantic Kernel and AutoGen to automate complex workflows.
- Applied deep learning and computer vision techniques to deliver scalable, real-world AI solutions across healthcare, travel, and industrial domains.
- Collaborated with cross-functional teams and contributed to technical documentation to support knowledge sharing and reproducibility.

HCL Technologies Ltd

Apr 2022 – Apr 2023

Software Engineer

Bangalore, Karnataka

- Designed, developed, and maintained Java-based applications and web services using Core Java and the Spring framework.
- Delivered customized software features aligned with client requirements, strengthening foundational skills in scalable, production-quality software engineering.

ANSA DECO GLASS PVT LTD

Mar 2021 – Aug 2021

Logistic Support

Surat, Gujarat

- Managed warehouse operations using SAP S/4HANA Cloud.
- Generated daily inventory reports for senior management decision-making.

Summary of Industrial Projects

Child Malnutrition Classification | Women Development and Child Welfare, Andhra Pradesh Govt. – AI/ML Developer

- Developed AI agents to automate verification and validation of malnutrition data for 1,648 children, eliminating manual errors and reducing validation time by 95%.
- Built a custom AI-enabled hybrid image annotation tool using 22 landmark points and contour extraction, processing 8,193 child left-arm images and cutting project development time by approximately 80%.
- Developed an offline, Android-based AI screening tool for Severe Acute Malnutrition (SAM) detection using 2D Geometric Morphometrics, enabling scalable field screening in low-resource environments.

- Achieved an overall accuracy of 90.68% using Linear Discriminant Analysis (LDA) on the 18-landmark model (removing redundant landmarks to avoid collinearity) after allometric correction, with fOPA aligned to the mean target.
- *Tech: Geometric Morphometrics, Linear Discriminant Analysis (LDA), tpsDig2, MorphoJ, MediaPipe, image segmentation.*

HCIVERSE – Multimodal No-Code AI Workflow Platform | AI Engineer

- Built a visual platform to design multimodal AI pipelines by connecting 30+ models across image, video, audio, and text.
- Developed with React Flow (frontend) and FastAPI (backend), featuring a custom DAG execution engine (Kahn's algorithm), real-time node streaming via WebSockets, secure authentication, and persistent project storage in MongoDB.
- Integrated Cloudflare Workers AI for asset caching and fallback routing.
- *Tech: Python, FastAPI, React Flow, Celery, Redis, MongoDB.*

AI-AGENTIC Chatbot – LogisEye (Dubai) | AI/ML Developer

- Built a multilingual AI chatbot with a custom UI to handle freight queries, shipment tracking, pricing insights, and more.
- Integrated voice, WhatsApp, and external APIs.
- *Tech: ChromaDB, Firecrawl API, Semantic Kernel, RAG (AWS Bedrock with Llama 3.1 405B).*

ML & Computer Vision-based Yarn Inspection – Shingora Textiles (Ludhiana) | AI/ML Developer

- Developed a data/image collection and image annotation pipeline API for a real-time defect detection system.
- *Tech: AWS EC2, MongoDB, S3, FastAPI, Python.*

AI Travel Trip Planner Assistant | AI/ML Developer

- Developed an agent-based travel assistant for personalized itineraries using conversational AI and real-time data.
- *Tech: Python, Semantic Kernel, Amadeus API, AutoGen, FastAPI, LLM (AWS Bedrock), FAISS DB.*

Poshan App – Anganwadi Services | Backend Developer

- Developed backend APIs for integration with mobile apps and a web dashboard for nutrition tracking and a face recognition-based attendance system.
- *Tech: Python (Flask), AWS (DynamoDB, Rekognition), S3, Lambda.*

Face Recognition for Furhat Social Robot – Event Check-ins | AI/ML Developer

- Developed a real-time face recognition system for guest authentication at events, achieving 96.8% accuracy.
- Enhanced recognition performance by integrating XGBoost with LBPH (+22% accuracy improvement), later extending to deep learning frameworks (MTCNN, DeepFace, MediaPipe) for robust deployment on the Furhat Social Robot.
- *Tech: MTCNN, DeepFace, MediaPipe, OpenCV, Python.*

Education

- | | |
|--|--|
| • Indian Institute of Technology (IIT) Jodhpur
Executive M.Tech in Artificial Intelligence | 2025 – 2027 (Pursuing)
Jodhpur, India |
| • Malwa Institute of Science & Technology
B.E. in Electronics & Communication Engineering (78.30%) | 2015 – 2019
Indore, M.P. |

Training & Certifications

- Orchestrate a Multi-Agent Solution using the Microsoft Agent Framework – Microsoft Learn, 2026.  Certificate
- Hands-on Workshop on Generative AI Agents & Workflow Automation – IIT Madras, June 2025.  Certificate
- Data Science Mentorship Program (DSMP) 2.0 – CampusX, May 2025.  Certificate
- Certificate in Python Programming with AI & ML – NIT Warangal, Oct–Nov 2021.  Certificate
- Fundamentals of Artificial Intelligence – IIT Guwahati (NPTEL), Jul–Oct 2021 (12 weeks).  Certificate

Extracurricular Activities & Achievements

- Selected as a Top Innovator and shortlisted for the prototype development phase – AI for Bharat Hackathon.  Certificate
- Served as Placement Coordinator & Team Leader during college, assisting in the management and coordination of campus placements, online recruitment tests, and system administration.  Certificate